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Tokenizing Industrial Assets: Multi-File Binding using an Average Perceptual Hash

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***Abstract*—The increasing digitalization in manufacturing and trends like the Industrial Metaverse drive the need for secure decentralized asset exchanges and industrial asset intellectual property (IP) rights protection. Utilizing distributed ledgers and non-fungible tokens (NFTs) linked by a hash enables the management and authentication of assets. However, challenges arise in authenticating the same asset's different versions and file formats using a single hash.**

In this work, we propose a new NFT structure that allows binding multiple files existing in different file formats, representing the same digital asset by combining a Merkle Tree (MT) with a calculated Average Perceptual Hash (APH). While the MT proves bitwise integrity, the APH verifies functional integrity for asset authentication. We evaluate our approach for exchanging Printed Circuit Board (PCB) designs in a decentralized ecosystem by calculating unique PCB fingerprints and the APH. Our results show a robust identification and integrity verification of PCB designs depending on the manipulation type. Our approach for asset authentication is generalizable to all asset classes with an appropriate perceptual hash.

***Index Terms*—Tokenization, Non-fungible tokens, Perpetual hash, Industrial Metaverse, Industrial Cyber-Physical Systems**

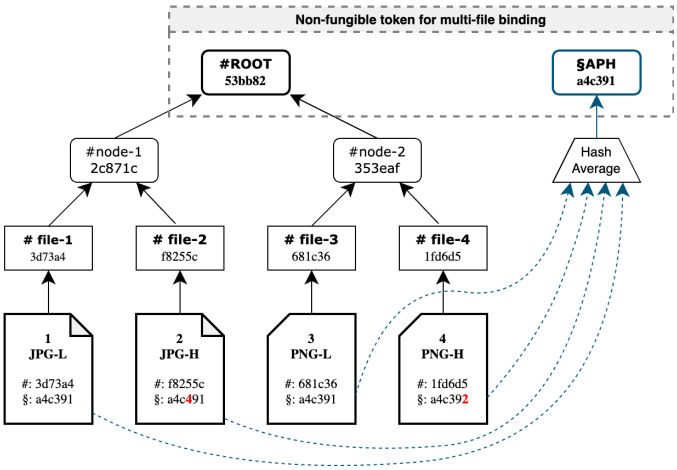


Fig. 1: Structure of the multi-file NFT for an image with four file representations. The files are JPG and PNG format combinations in low and high resolution. The MT root-hash #ROOT is constructed from the file's cryptographic hash values (marked with #), while the perceptual hash §APH is the average of the individual perceptual hashes (marked with §). Note how the perceptual hashes of files 2 and 4 slightly differ but are highly similar.

I. Introduction

Tokenized assets represented as NFTs are a new asset class within blockchain-based ecosystems [1] binding a physical asset to a cryptographic token. The bound token permits the owner to access specific content such as manufacturing instructions or digital twins, or other digital representations. The decentralized exchange of industrial assets is a cornerstone of the Industrial Metaverse ecosystem [2] that has a high demand for secure content exchange. Asset tokenization shows a surging demand with an estimated locked value of $2.15 billion as of May 2023 [3] on permissionless blockchains and has benefits over traditional asset management, such as better investor accessibility, revenue sharing, and frictionless cross-border transfers. Verifying digital ownership is a crucial NFT use case in the industry. Assets bound to tokens enable decentralized intellectual property (IP) rights management [4]. However, IP theft continues to be a tremendous problem, with an estimated $600 billion loss annually in the US [5]. IP protection is even more problematic for digital files in multiple formats shared with parties across the toolchain and manufacturing process. Furthermore, files can have different formats or levels of detail, e.g., a high-resolution image or a 3D model with different scales, and it must still be authenticated without additional information besides the linked APH. Therefore, they can be bound to an NFT representing the asset ownership, helping counterfeit the owner's IP rights.

This paper tackles the problem of verifying file integrity and ownership of tokenized assets, which may be encoded and stored in different file formats and versions but are exchanged and authenticated as a single NFT. A method to generate fingerprints for Printed Circuit Boards [6] was introduced and used to verify the process and perceptual hash performance against tampering. While various NFT-based counterfeit applications have been explored in [7], [8], we aim to extend tokenized assets for industrial asset exchanges that require different file formats and versions.

While the mere ownership of an NFT can be verified easily, proving that a physical asset or digital file version is associated with a token is more complex. Furthermore, extensive data is typically not saved directly on the blockchain due to high costs, so the only link between the data and token consists of a URI to an external data source. The challenge lies in how diverse file formats encoding identical content can be linked to a single NFT representing digital assets. Additionally, when multiple file representations are considered legitimate, an efficient method is required to determine if a given file is

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an authentic representation or a counterfeit with only minor changes. Our approach aims to solve this by introducing the APH in addition to the MT, creating a link between the physical asset and the token.

A. Contributions

As illustrated in Figure 1, our method binds multiple files to a single NFT by combining an *APH* with bitwise integrity provided through a cryptographic MT. This approach can be generalized to bind multiple file versions of all asset classes reducible to a perceptual hash. Additionally, we discuss and evaluate our approach through the following:

- Fingerprint generation for PCB designs (Section III-A) from the exported Gerber files [9] to create an Average Perceptual Hash for asset authentication.
- Evaluation of the APH robustness against general image tampering (Section IV-B, Section V-B) and manipulations of the PCB layouts such as changes of wire thickness and routing (Section V-D).
- APH performance for grouping slightly updated PCB versions into one multi-NFT compared to a standard perceptual hash (Section V-A).

II. Related work

This section gives context on PCBs and their evaluation for authenticating industrial assets in decentralized environments and covers related work about NFT-based IP protection solutions.

A. NFTs for IP protection and asset authentication

NFTs are increasingly used to manage and monitor asset life cycles and licenses [8]. Gupta [10] examined NFTs' role in addressing cybersecurity risks, counterfeiting, and piracy in additive manufacturing. Decentralized additive manufacturing networks represent a significant shift in industrial production and supply chain management [11]. These networks enable cost-effective, customized product creation and regional on-demand production, allowing them to adapt quickly to changing demands, minimize resource consumption, and promote a more agile, sustainable production ecosystem. Users of new decentralized platforms, as simplified in Figure 5, require methods to verify authenticity and ownership, which are essential for value creation and new business models. Immutable NFTs enable secure exchanges without intermediaries. However, linking physical assets with virtual tokens is challenging, often relying on simple resource identification in smart contracts, which leads to issues with external storage and file modifications. The APH is closing this gap by linking multiple files dynamically under one hash.

B. Perceptual Hashes

A perceptual hash is a fingerprinting technique that captures the perceived features of a multimedia file [12] instead of the raw data. Like a cryptographic hash, it outputs a short, fixed-length string of bits for a larger input. However, perceptual hashes should be robust against slight modifications in the input file that do not alter the perception of the file. For example, if the color of a single pixel in an image is slightly changed, the perception of the image will be unchanged. The perceptual hash value will remain the same or highly similar. By contrast, the altered pixel would cause a cryptographic hash value to change roughly half of all output bits, as it is intended to secure the exact integrity of a file. Thus, perceptual hashing enables efficient matching of multiple files to the NFT that holds the digital asset.

C. Printed Circuit Board and Gerber format

PCBs are an essential industrial asset used in decentralised additive manufacturing, involving different unknown parties that must exchange IP-relevant data. PCBs are a fundamental component of all modern smartphones, industrial machinery, and medical equipment with an expected market size of $100 billion by 2026 [13]. Following that, we chose them as a suitable asset class for the APH evaluation. PCBs consist of non-conductive materials like fiberglass and stacked coated copper layers. The layout is exported to a 2D vector Gerber format defined in the RS-274X standard [9] for the manufacturing process. The Gerber export is the base for our unique fingerprint generation and tokenization.

III. Multi-file binding to NFTs

Our method binds multiple files representing the same digital asset to a single NFT as illustrated in Figure 1. The files can differ in format, compression, or content resolution. The binding is achieved in two steps: first, by connecting the cryptographic hash values of the individual files using an MT. The MT allows efficient integrity and ownership verification of every file by combining all hashes bottom up until the root is reached. Second, since a cryptographic hash only recognizes exact bit patterns, the APH of all files is used to recognize similar files supplementing the root hash. Accordingly to Figure 1, the calculated MT root and APH are saved into the NFT smart contract.

A. Perceptual Hash requirements for PCBs

In our evaluation of the multi-NFT, the perceptual hash method used for the APH needs to reliably detect different file resolutions and formats of an asset while being robust against typical image tampering attacks, including alterations in color or brightness. The hash output must not exceed 32 bytes to fit into a smart contract string variable with minimal costs. Smaller hash sizes (E.g., 4 bytes) are more tolerant against tampering but lose distinctiveness. Furthermore, the calculation must be efficiently reproducible by every ecosystem participant.

We used a dataset of PCB layouts to verify the perceptual hash method [1]. A PCB design consists of a wire routing layout and vents forming electrical connections between components, creating a visually distinct pattern on the copper layer. This pattern captures the features reflecting the board's

[1]https://github.com/dangerousprototypes/dangerous-prototypes-open-hardware

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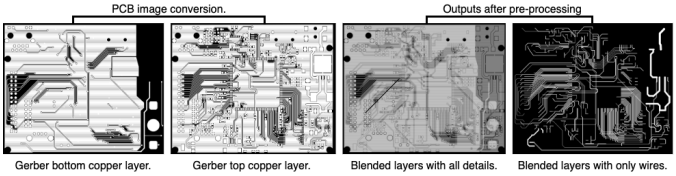


Fig. 2: The left side shows the exported top and bottom gerber copper layers of a two sided PCB design *Linux_board.va.brd* after image conversion to a PNG file. The right side shows the blended layers for the perceptual hash input after the preprocessing steps.

functionality when used to generate a fingerprint based on the PCB's copper layers. An image's DCT-based perceptual fingerprint, or analogously hash, must resist minor design manipulations that do not impact the PCB's intended functionality, such as slightly altering the position of a wire or another component, and help in the last data validity decision.

B. APH creation and image preprocessing

When authenticating industrial assets, the input data is not tied to a specific dataset, and the method used must be efficient on arbitrary data without training. Typically, a first preprocessing step reduces initial information and reinforces inherent features. For images, this can be done by converting all pixels to greyscale with 256 bits and downsampling the image to a square to resist uniform scaling, being resolutions agnostic. While the image's color information is discarded, the hash must resist brightness changes to distinguish blended layers out of the preprocessing. Furthermore, a longer hash captures more details and is more sensitive to minor changes.

Discrete Cosine Transform (DCT) based hash functions can robustly extract image features through the resistance of low-frequency DCT coefficients [12]. The DCT is applied to rows and columns of the standardized image's pixels array. The low-frequency DCT coefficients are then compared to their average, outputting a boolean array. We compute perceptual hashes for all files associated with the NFT to create the APH. The APH is derived by taking the median of each bit. The result is stored in the smart contract for similarity matching and as a universal fingerprint for all files. The APH can effectively group files of different formats, resolutions, and even slightly varied image versions to detect counterfeits and prove authenticity.

IV. Evaluation

This section provides an overview of PCB preprocessing steps to extract relevant features from Gerber files to generate the APH. We likewise verify the APH against tampered images and direct manipulations of PCB layout elements, such as wire thickness and vent positions. PCBs are a highly relevant asset class for evaluating our multi-NFT approach, but the method applies to any asset reducible to an APH. The dataset used contains 95 open hardware PCB designs. Despite its small size, this dataset is sufficient to evaluate the basic feasibility of the multi-NFT concept and general APH performance.

Standard PCBs primarily consist of two copper layers, top and bottom, electrically isolated and connected through vias. Boards with more layers offer better electrical isolation but get more complex and costly to produce. Our fingerprint method extracts information from all copper layers and blends them into one output image with different grayscale values. PCBs vary in size and shape to fit specific devices. The wire layout forms the electrical connections and creates a visually distinctive pattern on the copper layer (see Figure 2). The wire thickness and routing layout directly influence electrical resistance and interference between components, as too close positioning can cause unwanted electromagnetic coupling and increased heat development [14]. Non-functional modifications to the routing layout or other components should still result in a similar perceptual hash. Since these manipulations are complex to auto-generate, we simulate actual tampering attempts by applying different manual manipulations to the PCB layouts. Section V-D overviews these manipulations, such as changes in wire thickness, routing layout, and component removal. Thus, a perceptual hash must capture the PCB's core features to support authentication.

A. PCB data extraction and image conversion

To validate the verification process against the APH, we first exported the Gerber files from the PCB designs and preprocessed layers into one blended fingerprint image. The CAE software Eagle 9.62 was used to export the manufacturing data and create a set of manipulated layouts, including images with only the wire routing.

The exported copper layers are processed with a Python script rendering the layers over each other in grey scale on a black background as shown in Figure 2. The layer order must be the same as for fabrication, with the bottom and top treated as individual layers. Manufacturers may support up to 20 layers for complex designs. Multiple layers can be added to the final blended image with brightness and transparency increments of the layers. The layers are merged with greyscale color in the rendering step from a vector to a pixel-based image format. The RGB values between 128-256 bits are applied to map the layers divided by the layer index to capture a distinguishable pattern containing all relevant features of the PCB, creating a unique fingerprint. Figure 2 shows the outcome of the converted gerber top and bottom copper layers after pre-processing to a PNG image into the layer blended version and extracted wires.

B. Creation of tampered images

Additional image instances with up and down-scaled resolutions and brightness are generated to test the APH robustness against fundamental tampering attacks. The calculated APH of the base and resolution images are compared against the tampered and original images. The applied image manipulations include:

- Scaled resolutions of 0.5x and 2x.
- Cropped images of 0.97x, 0.98x and 0.99x.
- Image rotation by 180°.

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TABLE I: The table shows the APH performance against image manipulations within single designs (left) and wires (right), including color manipulations and transformations. The APH was calculated from the original design and scaled resolutions. While resistance to color and size manipulations is strong, partial structural changes significantly impact the normalized Hamming distance (nHD). Additionally, the results indicate that extracting only the PCB's wires, without other components, is sufficient to generate a distinctive fingerprint.

L	MPL	HD		nHD (%)		MAD		τ > 90%	
8	Original	0.15	0.2	99.06	98.75	0	0	100	90
	Bigger	0.15	0	99.06	100	0	0	100	100
	Smaller	0.15	0	99.06	100	0	0	100	100
	Brighter	1.95	2	87.81	87.5	0	0	25	10
	Darker	0.38	0.6	97.60	96.25	0	0	88	70
	Flipped	12.2	13.6	23.96	15	2	2	0	0
	Tile 90°	0.48	0.5	96.88	96.88	0	0	87	80
16	Original	0.53	0.3	99.17	99.53	0	0	100	100
	Bigger	0.53	0.7	99.17	98.91	0	0	100	100
	Smaller	0.53	0.5	99.17	99.22	0	0	100	100
	Brighter	9.32	10.7	85.44	83.28	2	1	20	20
	Darker	0.93	0.9	98.54	98.59	1	0	100	100
	Flipped	50.2	54.9	21.59	17.03	6	6	0	0
	Tile 90°	4.3	5.3	93.28	91.72	2	3	78	70
32	Original	3.18	0.9	98.76	99.65	1	0	97	100
	Bigger	3.2	0.7	98.75	99.73	1	0	97	100
	Smaller	3.18	1.1	98.76	99.57	1	0	97	100
	Brighter	41.3	40.7	83.85	84.10	7	9	3	10
	Darker	5.03	6.5	98.03	97.46	2	2	97	100
	Flipped	203	220	20.67	13.87	16	19	0	0
	Tile 90°	34.6	28.2	86.48	88.98	14	12	35	50

- 90° rotation of a tile with the size 0.1x in the center.
- Image enhancement with 0.5 (darker) and 1.5 (brighter).

An arbitrary threshold τ can be chosen for the tamper sensitivity of an APH comparison. The resulting *normalized hamming distance* must be $>= \tau$ to count as a valid match. For evaluating the individual PCBs, we chose a τ of 90% and 80% for the groups to allow more minor differences. A PCB's wire and layer thickness directly affects its heat and power management, as defined in IPC-2221 and IPC-2152 [14]. Following that, significant changes to the wire's thickness and positions that are too close to other components may render the PCB unfunctional. Slight tampering of changes that do *not* affect functionality is, therefore, still a potential attack vector and needs to be detected. Which is tested with manually generated PCB design manipulations:

- Removal of small sub-parts such as vents or short wires.
- Reduction of the wire thickness by 2/mil each step.
- Wire routing changes with the same functionality.
- Moving components on the board layout in steps of 0.35mil, 3.5mil, and 50mil.

C. Evaluation setup

We evaluated the APH efficiency with multiple hash sizes 4, 8, 16, and 32. The base perceptual hash configuration is described in Section III-A. The APH was calculated for all PCB designs after converting the exported Gerber copper to layers to a blended image, based on the preprocessing shown in see Figure 2. Different resolutions and tampered versions were created for all images to test tampering. Additionally, one set of 34 PCBs was created, grouping identical PCBs with minor updates to verify the APH tolerance for design changes. The PCB set of single designs consists of 95 blended copper layers, including wires, vents, custom components, and one with only the extracted wires.

V. Average Perceptual Hash Results

This section provides a detailed overview of the APH performance for detecting instances with different resolutions and versions. Additionally, the APH is tested against tampered layouts and malicious manipulations of the PCB. The tests ran with APH sizes (4, 8, 16, and 32) and measured the nHD similarity and the dataset's Median absolute deviation (MAD).

A. APH performance for grouping different versions to the NFT

Besides linking multiple file formats and resolutions to one token, the APH allows grouping multiple versions of a layout as long as it has only minor changes, e.g., Figure 3 has slightly modified vent size and wire routing changes marked by the red areas. In Table II, the grouped designs were tested against their generated APH with a threshold of 80%. Furthermore, the comparison of plain perceptual hash was added as a baseline. The APH performed moderately better for 4-16 hash lengths. While grouping different layout versions is a valid option and performs best with smaller hashes, the validity threshold and size of the image diff is a trade-off that the NFT creator can maintain. The grouped design set showed no collisions like the other datasets, enabling the grouping of different versions to a token.

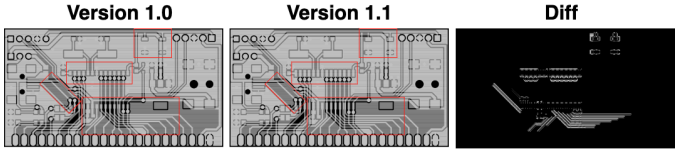


Fig. 3: Shows two PCB versions differing in the routing layout. The white area on the right side marks the changes. All images generated from the PCB layout "pic18f2550" reached a similarity of 87.5% against their mutual APH and can be grouped into one multi-NFT. The changed area is relatively hard to spot by the bare eye, but the "diff" result shows that a significant part is affected.

TABLE II: Performance of the APH and the plain perceptual hash for the grouped PCB versions.

L	HD	APH (%)	pHash (%)	MAD	τ > 80%
4	0.1	94.74	93.68	0	79
8	1.43	93.75	91.89	1	100
16	7.29	91.61	90.23	4	84
32	32.8	87.81	86.62	24	79

B. Cropping

All fingerprint images were pre-cropped 5% of their size to remove the PCB's empty safe area for fabrication. Cropping 1% in the reduced area significantly impacted the APH hash performance. APH-8 is less affected, with 52% valid matching, and APH-16 still has 32% valid matching. The APH-32 becomes completely invalid. The cropping vulnerability is, in most cases, acceptable, as the removal may render the PCB non-functional. A cropping-resistant image hash was introduced by

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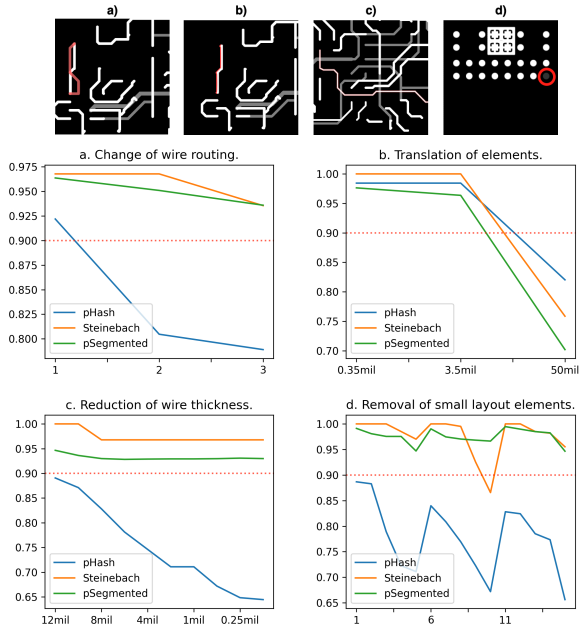


Fig. 4: The plots depict comparisons between the APH based on pHash with a size of 32 bytes, a segmented version of the pHash by splitting up the pixel array into multiple cells, and Steinebach's cropping-resistant hash. Data points above the red line are deemed valid. Exemplary modifications are illustrated in the top row of four images, with the tamperings highlighted in red.

Steinebach [15], which detects all cropped images with 100% accuracy but therefore was complex to configure for detecting fine-grained details tamperings such as wire thickness simultaneously (See Figure 4). Furthermore, cropping resistance may help detect individual functional groups on a PCB. For example, if a PCB is battery-powered, an attacker could crop away the charging circuit and reupload the remaining layout as counterfeit.

C. False positives and collisions

The APH showed no false validity over 90% and a throughout low HD for the tested PCBs. The APH-4 captures less information and leads to a minor increase in the average similarity of around 2.5% on all image sets. Hence, a hash size greater than four may be recommended to get collision-resistant results.

D. Gerber Layout Modifications

The PCB layout can be changed by modifying vent positions, wire routes, and the wire's thickness (see Table III). As closely routed wires can have interferences [14] and wire thickness is tied to the electrical performance, wires can only be modified to some extent without affecting the functionality. Wire thickness and distances are measured in mil, which refers to 0.001 inches. The PCB layout modifications were tested against the APH, and results are shown in Figure 4 that include no-functional change resistance, random elements removal, and minor wire changes.

TABLE III: Changes of the APH when reducing the wire thickness. The reduction was done in four mil steps and strongly affected larger hashes.

L	mil	HD	nHD (%)	MAD	$\tau > 90\%$
8	2	0.9	94.38	0	70
	6	1.7	89.38	1	60
	10	2.7	83.12	2	30
	14	3.1	80.62	2	30
16	2	4.4	93.12	0	70
	6	11.1	82.66	1	60
	10	14.3	77.66	2	30
	14	16.2	74.69	2	30
32	2	24.5	90.43	11	70
	6	57.3	77.62	15	20
	10	73.6	71.25	29	10
	14	90.9	64.49	42	10

VI. Discussion

Protecting digital IP can be challenging. Similar to the software piracy dilemma, unlicensed software use still accounts for 37% of installations globally [16] cycling between new copyright protection mechanisms and their malicious workarounds. Distributed Ledgers may provide an alternative solution by enabling tamper-proof digital ownership. Our NFT structure shown in 1 can bind different file versions together and prove their authentic ownership to a third party through the APH and MT. The evaluation of the APH demonstrated the successful manipulation detection of the PCB layouts based on generated fingerprints of the PCB's Gerber files. Therefore, any ecosystem participant can reproduce the PCB's fingerprint and verify its authenticity.

A. Limitations of the approach and further research

We compared different configurations for the APH. The perceptual hash can be chosen depending on the use case. Based on a DCT (pHash), we compared 4 to 32 bytes lengths and two image preprocessing modes, aspect and stretch. The high-frequency factor was set to six, while a higher factor leads to more captured details and change sensitivity. The results of Figure 4 show the comparison of a plain perceptual hash to a segmented and cropping-resistant hash. While the plain hash performs less accurately, the segmented hashes often need to reflect the increasing change. However, the correlation between manipulation and the effect on the actual features by moving, removing, and modifying the thickness must be more evident to generalize along with a larger PCB dataset and needs further research. Additionally, the segmented hashes have not yet been converted to an average perceptual hash, which still needs to be evaluated.

Tampering attacks, such as inverting color or flipping, make the image unrecognizable, which can be problematic, e.g., when one color specifies the cutout area in the fabrication process. Additional steps should be applied in a production environment for authentication, such as calculating multiple hash variants of all image directions and an inverted version. Table I shows low values under 20% for brightness manipulations related to the image preprocessing step where all Gerber layers are blended into one image with different grey-scaled RGB values between 128 - 255. A substantial increase in

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brightness reduces the differentiability. Therefore, fingerprint generation should be done using the original files through the validator and not be passed. Another result is the low detection rate of rotating single components on the board or removing more significant parts without segmentation. Furthermore, the fabrication process of a PCB design can be done in batches with multiple copies in one Gerber file, which adds much noise. The perceptual base algorithm must provide more results for this case, where a batch of two boards already resulted in 18% similarity. An improvement may be introducing image segmentation as suggested by Steinebach [15] together with the APH and a calibrated change sensitivity.

B. Generalization

The proposed NFT structure binds 2D images to a token. The method is not limited to pictures but can be generalized to any media format, reducible to a perpetual hash. Future work is needed to evaluate a broader range of assets, such as generating perceptual hashes for 3D computer-aided design. A 3D fingerprint generated from a CAD file could be linked to the token and secured against mesh tampering [17]. Other asset classes like text, audio, or video may function accordingly.

C. NFT-based IP authentication for decentralized applications

Tokenization links a physical asset to a digital token on a DLT, representing ownership, access rights, or any certificate. The tokenized PCB is immutably deployed and stored via a smart contract. The exported manufacturing data, including the bill of materials, drilling instructions, and other fabrication metadata, can be added to the asset contract through an IPFS storage with the IPFS folder hash saved to the contract, together with the MT root for the cryptographical file integrity and the APH. Minted tokens can be exchanged and used to authenticate PCB designs in the proposed three-way decentralized ecosystem. Figure 5 shows the simplified flow of adding and exchanging authenticated assets. Designers can upload new assets to the platform, proving authorship without intermediaries, while buyers verify authenticity and proof ownership for fabricators.

VII. Conclusion

This paper presented a novel method of binding multiple versions of an asset, differentiating in file formats and resolutions to a token by combining a generated APH with the cryptographic hashes of a Merkle Tree. The APH generation process was evaluated against 95 functional PCB designs through their exported manufacturing data in Gerber format. In a preprocessing step, the Gerber's copper layers were blended into a unique fingerprint for an APH that can be easily stored and compared. Different fingerprints were tested based on the plain copper layers, one with only wires, and a set of grouped PCB designs consisting of versions with minor layout updates. The APH was tested against manipulations of the output image, such as brightness, size, transformations, and cropping. Furthermore, modifications in the PCB designs, like rerouting wires and moving vents, were evaluated and verified based on a plain and segmented perceptual hash method and the detection of wire thickness changes. We discussed using multi-file binding NFTs to exchange industrial assets in a trustless, decentralized ecosystem. Assets can be exchanged through a marketplace smart contract and authenticated for different file representations. In conclusion, the NFTs provide new ways of managing IP and exchanging PCB designs in decentralized applications. The APH showed a slightly improved performance compared to a single perceptual hash. Furthermore, the APH provides a reproducible link between the exported Gerber files and ownership of the tokenized PCB, proving it can detect the most common image and layout tampering.

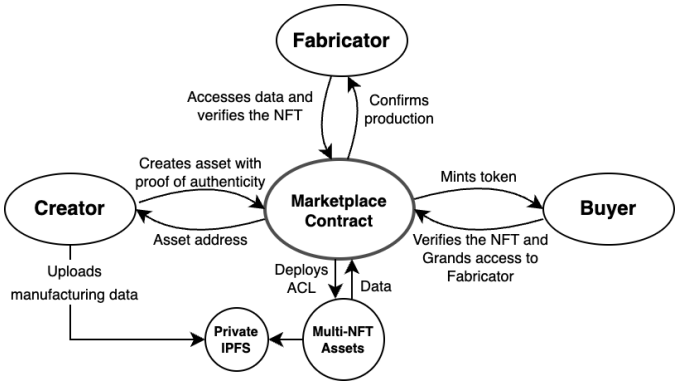


Fig. 5: The general flow of a decentralized IP-protecting, three-way smart manufacturing system that uses multi-NFTs to authenticate assets and provides secure access to critical fabrication data without a middleman.

References

[1] R. Heines, C. Dick, C. Pohle, and R. Jung, *The Tokenization of Everything: Towards a Framework for Understanding the Potentials of Tokenized Assets.*

[2] M. Prummer, E. Regnath, S. Singh, and H. Kosch, "From virtual worlds to real-world impact: An industrial metaverse survey," in *Future of Information and Communication Conference.* Springer, 2024, pp. 592–613.

[3] F. Carapella, G. Chuan, J. Gerszten, C. Hunter, and N. Swem, "Tokenization: Overview and financial stability implications," no. 2023, pp. 1–29.

[4] E. Lee, "NFTs as decentralized intellectual property."

[5] D. Blair, "The commission on the theft of american intellectual property."

[6] M. Montrose, *Printed circuit board design techniques for EMC compliance.* IEEE press Piscataway, NJ, 1996, vol. 1.

[7] R. Mehta, N. Kapoor, S. Sourav, and R. Shorey, "Decentralised image sharing and copyright protection using blockchain and perceptual hashes."

[8] S. M. H. Bamakan, "Patents and intellectual property assets as non-fungible tokens; key technologies and challenges," vol. 12.

[9] "The gerber layer format specification," Ucamco NV, Tech. Rep., 2023.

[10] N. Gupta, "ObfusCADe: Obfuscating additive manufacturing CAD models against counterfeiting: Invited." ACM.

[11] M. T. R. Insights, "The emergent industrial metaverse," MIT, Tech. Rep., 2023.

[12] A. Neelima and K. M. Singh, "A short survey on perceptual hash function."

[13] P. analysis, "Printed circuit report," 2022.

[14] D. J. Adam, "Strombelastbarkeit von leiterbahnen."

[15] M. Steinebach, "Efficient cropping-resistant robust image hashing," in *2014 Ninth International Conference on Availability, Reliability and Security.*

[16] "Software management: Security imperative, business opportunity," BSA, Tech. Rep., 2018.

[17] M. Prummer, E. Regnath, and H. Kosch, "Onion-hash: A compact and robust 3d perceptual hash for asset authentication," *Computer-Aided Design*, vol. 175, p. 103752, 2024.

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